Accenture — Aptitude Test 2

Questions 26–50 of 100 · 25 questions

Question 26. In a race of 200 m, A can beat B by 31 m and C by 18 m. In a race of 350 m, C will beat B by what distance?

Solution:
Distance travelled by B and C when A completes the race = 169 and 182
Ratio between distance of B and C = 13 : 14
Assume that ‘d’ is the distance travelled by B when C completes the race.
Then,
13/14 = d/200
d = 185.7
C beats B by 14.3 meters.

A c c e n t u r e

Question 27. 729ml of a mixture contains milk and water in ratio 7:2. How much water should be added to get a new mixture containing half milk and half water?

Solution:
Quantity of Milk = (7/9) x 729 = 567
Quantity of water = (2/9) x 729 = 162
To have half water and half milk, the quantity of water must be increased to 567.
Available quantity of water = 162.
Required quantity of water = 567 – 162 = 324.

The quantity of mixture which is 198 liters less than the actual mixture = N – 198
According to the question, percentage of petrol in the less quantity mixture – percentage of petrol in the actual mixture = 13.33%
So,
[99/(N-198)]x100-[99/(N)]x100=13.33
By solving the above equation, we get
99N – 99N
N = 495
Concentration of Petrol in the actual quantity = (99/495) x 100 = 20%

Question 28. The proportion of milk and water in 3 samples is 2:1, 3:2 and 5:3. A mixture comprising of equal quantities of all 3 samples is made. The proportion of milk and water in the mixture is?

Question 29. In a mixture of petrol and kerosene, petrol is only 99 liters. If this same quantity of petrol would be presented in another mixture of petrol and kerosene where total volume would be 198 liters less than the actual mixture then the concentration of petrol In the present mixture would have been 13.33% point less than that. What is the concentration of petrol in actual mixture?

A c c e n t u r e

Solution:
Let us Assume that the actual quantity of the mixture = N

Question 30. An alloy of zinc and copper contains the metals in the ratio 5 : 3. The quantity of zinc to be added to 6 kg of the alloy so that the ratio of the metal may be 3 : 1 is:

Solution:
There are 2+1 = 3 parts in 1st mixture, 5 parts in 2nd mixture and 8 parts in 3rd

mixture. To make easy calculation, let us assume that the quantity taken in each mixture is the LCM of (3,5,8).
Quantity taken in each mixture = 120
Mixture 1:
Milk : Water = 2 : 1 = 80 : 40
Mixture 2:
Milk : Water = 3 : 2 = 72 : 48
Mixture 3:
Milk : Water = 5 : 3 = 75 : 45
After mixing all the mixtures,
Final quantity of Milk
= 80 + 72 + 75 = 227
Final quantity of Water
= 40 + 48 + 45 = 133
Final ratio = 227 : 133

Solution:
Quantity of zinc in 6 kg = (5/8)6 = 3.75
Quantity of copper = 6 – 3.75 = 2.25
For the quantity of zinc to be thrice that of the quantity of copper, zinc's weight should be 3 x 2.25 = 6.75 kg
Available quantity of zinc = 3.75 kg

Quantity of zinc to be added
= 6.75 – 3.75 = 3 kg.

Question 31. A mixture of 40 liters of milk and water contains 10% water. How much water should be added to this so that water may be 20% in the new mixture?

Solution:
Quantity of water in original mixture
= (10/100) x 40 = 4 liters.
Quantity of milk in original mixture = 36
After adding water to the mixture, the new mixture has 20% water, that means the percentage of milk in the new solution is = 80%.
The quantity of milk is same in both mixtures.
So, 80% of New quantity = 36
(80/100) x New quantity = 36
New quantity = 45 liters
The quantity of water added
= 45 – 40 = 5 liters.

Question 32. Three jackals Paar, Maar and Taar together have 675 loaves of bread. Paar has got three times as much as Maar and 25 loaves more than Taar. How many does Taar have?

Solution:
Let the number of loaves with Paar, Maar and Taar be P, M and T respectively.
P + M + T = 675 ---(1)
P = 3M ■M = P/3
P = T + 25 ■T = P – 25
Substitute M and T in (1)
P + (P/3) + P – 25 = 675
7P/3 = 700
P = 700 x 3/7
P = 300
T = P – 25 = 300 – 25 = 275
The number of loaves with T = 275

Question 33. A box contains 90 bolts each of 100 grams and 100 bolts each of 150 grams.

If the entire box weighs 35.5 kg, then what is the weight of the empty box?

Solution:
Let Weight of Box be B.
B + (90x100) + (100x150) = 35500gm.
(Express both sides in terms of grams).
B + 9000 +15000 = 35500
B = 11500 grams
Weight of the box is 11.5 kg

Question 34. Ravi had got twice as much marks as Ramu. His teacher had made him a promise that, for every mark he scores above Ramu, he would be awarded 50% of those marks as bonus. Find the ratio of his bonus marks to the total marks of Ravi and Ramu.

Solution:
Assume that the mark scored by Ramu is N.
So, the mark scored by Ravi = 2N
Marks scored by Ravi more than Ramu = 2N – N = N
Bonus marks awarded to Ravi = N/2
Required ratio = (N/2) : (3N) = 1 : 6

Question 35. When Raja was born, his father was 32 years older than his brother and his mother was 25 years older than his sister. If Raja's brother is 6 years older than Raja and his mother is 3 years younger to his father, how old was Raja's sister when Raja was born?

Solution:
When Raja was born, age of his brother is 6 years.
Father's age at that time = 6 + 32 = 38.
Age of mother at that time = 38 – 3 = 35
Age of sister is 25 less than age of mother.
Age of sister when Raja was born = 35 – 25 = 10

Question 36. A father is three times as old as his son. After fifteen years the father will be twice as old as his son's age at that time. Hence the father's present age is?

Solution:
Assume that the father's present age = F

Son’s present age = S
At present, F = 3S
After 15 years,
F + 15 = 2(S + 15)
F = 2S + 15
Substitute F = 2S + 15 in F = 3S
2S + 15 = 3S ■S = 15 and F = 45

Question 37. Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11:10. What is Sagar's age at present?

Solution:
Kunal's present age = K
Sagar’s present age = S
From the question, age of Kunal and
Sagar 6 years ago = 6x, 5x
Four years hence, the ages = 11x, 10x
Difference in ratio = (11 – 6) or (10 – 5)
= 5
Difference in years = 10
x = diff in years/diff in ratio ■x = 2.
Age of Sagar 6 years ago = 5(2) = 10
Sagar’s present age = 16.

A c c e n t u r e

Question 38. The age of the grand father is the sum of the ages of his three grandsons. The second grandson is 2 years younger than first one and the third one is 2 years younger than the second one. Then what will be the age of the grandfather with respect to the second grandson?

Solution:
Total marks in 10 papers
= 80 x 10 = 800
Removing the highest and the lowest
score the average is = 81.
Total marks of the remaining 8 subjects
= 8 x 81 = 648.
Total
marks
scored
in
highest
and
lowest = 800 – 648 = 152.
Highest + Lowest = 152
92 + Lowest = 152
Lowest = 152 – 92 = 60
His lowest score is 60.

Solution:
Let the age of first grandson is x.
Agee of second grandson = x – 2.
Age of third grandson = x – 4.
Age of grandfather
= x + (x – 2) + (x – 4) = 3x – 6
Ratio of age of grandfather to age of second grandson
= (3x – 6) : (x – 2)
= 3 : 1
Grandfather's age is thrice that of the second grandson's age.

Question 39. Total of the ages of A, B and C at present is 90 years. Ten years ago, the ratio of their ages was 1: 2: 3. What is the age of B at present?

Solution:
Total age of A, B and C 10 years ago
= 90 – 3(10) = 60
The ratio of their ages 10 years ago
= 1 : 2 : 3
Age of B 10 years ago = (2/6) x 60 = 20
Present age of B = 20 + 10 = 30 years.

Solution:
Total marks of five persons
= 39 x 5 = 195
Saheep = Ramesh + 7
Ramesh = Ajith – 9
So, Saheep = Ajith – 9 + 7 = Ajith – 2
Sachu = Saheep + Ramesh
Sachu = Ajith – 2 + Ajith – 9 = 2Ajith-11
Sachu + Karna = 110
2Ajith – 11 + Karna = 110
2 x 32 – 11 + Karna = 110
Karna = 110 + 11 – 64
Karna = 57

Question 40. The average age of 10 members of a committee is the same as it was 4 years ago, because an old member has been replaced by a young member. Find how much younger is the new member?

Solution:
Let the average age of 10 members 10 years ago = N
Total age 10 years ago = 10N.
Total age at present before replacing
= 10N + 4 x (10) = 10N + 40
Present total after replacing = 10N
Difference between the age of person remove and the person replaced
= 10N + 40 – 10N = 40 years.

Question 41. The average marks of a student in ten papers are 80. If the highest and the lowest scores are not considered, the average is 81. If his highest score is 92, find the lowest.

Question 42. In an exam, Ajith, Sachu, Karna, Saheep and Ramesh scored an average of 39 marks. Saheep scored 7 marks more than Ramesh. Ramesh scored 9 fewer than Ajith. Sachu scored as many as Saheep and Ramesh scored. Sachu and Karna scored 110 marks them. If Ajith scores 32 marks then how many marks did Karna score?

Question 43. The average weight of a class of 24 students is 36kg. When the weight of the teacher is also included, the average weight increases by 1kg. What is the weight of the teacher?

Solution:
Total weight of 24 students
= 36 x 24 = 864
Total weight of 24 students and 1
teacher = (24 + 1)(36 + 1)
= 925
Weight of teacher = 925 – 864 = 61 kg.

Question 44. Average temperature on Wednesday, Friday and Thursday was 26o. Average temperature on Thursday, Friday and Saturday was 24o. If the temperature on Saturday was 27o, what was the temperature on Wednesday?

Solution:
Wed + Thu + Fri = 3 x 26 = 78o

Thu + Fri + Sat = 3 x 24 = 72o

Thu + Fri + 27 = 72o

Thu + Fri = 45o

Substitute Thu + Fri = 45 in
Wed + Thu + Fri = 78o

Wed + 45 = 78
Wed = 78 – 45 = 33o C

Question 45. A certain quantity of 40% solution is replaced with 25% solution such that the new concentration is 35%. What is the fraction of the solution that was replaced?

Solution:
Assume that the quantity of the mixture is 100 liters.
In 100 liters, 40 liters was acid.
If 1 liter of 40% solution is replaced with 1 liter of 25% solution, we will have
40 – 0.4 + 0.25 = 39.85 liters of acid.
For every 1 liter replacement, the percentage decrease = 0.15%
For 5% decrease the quantity to be replaced = 5/0.15 = 33.33 liters.
Fraction of 33.33 in 100 liters
= 33.33/100 = 1/3

Question 46. If a shopkeeper accidentally sells a pen at double its actual selling price, his profit increases 4 fold. Then he realizes his mistake and sells other pens at their original selling price.

Find his actual profit percentage.

Solution:
Assume that the actual selling price = S
and the cost price = C
Profit = S – C
At double the selling price, the profit is 2S – C.
Given, 2S – C = 4(S – C)
4S – 2S = 4C – C
2S = 3C
S/C = 3/2
This implies, for cost price = 2, the selling price = 3
Profit percentage = 50%.

Question 47. A ball dropped from H height and moves 80% of height each time. Total distance covered is?

Solution:
Distance in 1st drop = H
Distance in 2nd drop = 2 x 0.8H = 1.6H

Distance in 3rd drop = 2 x 0.8 x 0.8 H

= 1.28H
Distance in 4th drop = 2 x 0.8 x 0.8 x0.8H

= 1.03H
By adding all the distance from 2nd drop
to the last drop, the maximum distance
the ball can jump = 8H
Total distance travelled = 8H + H = 9H.

Question 48. A student gets an aggregate of 60% marks in five subjects in the ratio 10 : 9 : 8 : 7 : 6. If the passing marks are 50% of the maximum marks and each subjects has the same maximum marks, in how many subjects did he pass the exam?

Solution:
Assume that the maximum marks in
each subject = 100
Total marks scored by the candidate
= (60/100) 500 = 300
Let the actual marks be 10x,9x,8x,7x,6x.
10x + 9x + 8x + 7x + 6x = 300
40x = 300 ■x = 7.5
Marks in each subject
= 75, 67.5, 60, 52.5, 45
He passed in 4 subjects.

A c c e n t u r e

Assume that the cost price of sunglass
is B.
Given,
(96/100)A + (104/100)B = 2600 and
(104/100)A + (96/100)B = 2700
From the above two equations we get
96A + 104B = 260000
---(1)
104A + 96B = 270000
---(2)
By solving (1) and (2)
A = 1950 and B = 700

Question 49. After a discount of 11.11%, a trader still makes a gain of 14.28%. At how many percent above the cost price does he mark his goods?

Solution:
Assume that the marked price = 9.
After 11.11% discount the price of the
product = 8
By selling Rs. 7 product for Rs. 8, one can
gain 14.28% profit.
Cost price : Marked price = 7 : 9
Marked price is 28.57% above cost price.

Question 50. A man purchased a watch for Rs. 400 and sold it at a gain of 20%. The selling price of the watch is?

Solution:
Assume that the price of gold was Rs.100
After increasing the price by 30%, the new price is = Rs. 130.
Rs. 30 worth of ornaments should be reduced.
Percentage decrease
= [(130 – 100)/130] x 100 = 23.07%

Solution:
Selling price = CP + (20/100)CP
Selling price = 400 + (20/100)400
Selling price = Rs. 480